

Confidential

产品开发更改

Doc. Approval: RBCD/EIS

Effective date: 20250714

DC No. / 开发更改编号: PDECR_0053

Date / 日期: 20251105

Step 1: Change request & Basic information

Customer project Name/ 客户项目名称:	57cc e-compressor	MCR No. / MCR号:	NA
Affect Product No. / 产品		Component No. / 零部件号:	
57cc e-compressor F01Z 7P0 013-04 CNHTC 57cc e-compressor F01Z 7P0 00H-01 XCMG 57cc e-compressor F01Z 7P0 012-04 Geely FD 57cc e-compressor F01Z 7P0 012 Geely SOP 57cc e-compressor F01Z 7P0 014-02 Qingling H47 45cc e-compressor F01Z 7P0 01E-01 Qingling 4.5T BEV		Offerdrawing: F01Z7D0013-01 F01Z7D000H-01 F01Z7D0012-01 F01Z7D0012 F01Z7D0014-01 F01Z7D001E-01	EOL function PMB: F01Z7S0003 F01Z7S0003-01
Sample type / 样品类型:	☐ A ☐ B ☐ C ☒ FD	Initiator/发起人:	ZHU Qiangmin(RBCD/EIS3)

Reason of changes / 更改理由:

- 改善拔出时保护帽冲击力。
- PMB符合生产实际。

Customer request / 客户要求 ☐	Supplier request / 供应商请求 ☐	Design optimization / 设计优化 ☐
Correction / 更正(错误) ☐	Process / 工艺 ☐	Other / 其它 ☐

Step 2: Change proposal & Change discription

Before Change

After Change

1. 更改Offerdrawing中关于压缩机出厂时内部充氮压力, 从0.15MPaA~0.25MPaA更改为0.13MPaA~0.19MPaA.

When the compressor leaves the factory, it is internally sealed with dry nitrogen gas of 0.15 MPaA~0.25 MPaA, and when it is switched on, it is discovered that the compressor has no positive pressure, and it should be notified to RB's quality Department for confirmation and use

压缩机出厂时内部密封0.15MPaA~0.25MPaA的干燥氮气, 开启使用时发现压缩机没有正压力, 应通知博世公司质量部门确认后使用

After removing the inlet and outlet protective caps, it should be

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2. 更改功能检测规范3.1.4测试点中关于进口温度变更, 从24.0±3°C改为25.0±6°C.

3.1.4 Test points/测试点

Test points 测试点	Setting value		Result specification			
	Set Speed 设定转速 [rpm]	Inlet pressure 进口压力 [barg]	Inlet temperature 进口温度 [°C]	Outlet pressure 出口压力 [barg]	DC Power current 直流电流 [A]	Rise time 升压时间
1	(1000±150)	(1.0±0.2)	(24.0±3)	(8.3±1.5)	(0.9±0.2)	Record

3.1.4 Test points/测试点

Test points 测试点	Setting value		Result specification			
	Set Speed 设定转速 [rpm]	Inlet pressure 进口压力 [barg]	Inlet temperature 进口温度 [°C]	Outlet pressure 出口压力 [barg]	DC Power current 直流电流 [A]	Rise time 升压时间
1	(1000±150)	(1.0±0.2)	(25.0±6)	(8.3±1.5)	(0.9±0.2)	Record

Step 3.1 Impact analysis / 影响分析

1. Function&Performance will be influenced? / 产品功能性能影响?

☐ yes/是 ☒ no/否

If Yes, product impact evaluation?
如果是, 产品影响评估? (如果是初次样件放行, 空白处请说明样件目的)

Whether need sample to test? Plan?
是否需要样件进行验证? 计划?

不需要

Testing / 签名 朱强敏

2.1 Interface&boundary to customer will be influenced? / 接口&边界是否影响客户?

☒ yes/是 ☐ no/否

Please explain the impact and whether it has been confirmed by the customer? 请说明影响和是否得到客户的确认?

offerdrawing充氮压力变更需告知客户;

Customer PM / 签名 孙世杰 钱树

2.2 Interface&boundary to internal will be influenced? / 接口&边界是否影响内部?

☒ yes/是 ☐ no/否

If Yes, internal impact evaluation?
如果是, 内部影响是否评估? (例如: 测试接头, 噪音测试夹具等)

1.测试线需要更改设定;
2.更改不同压力下保护帽弹出状态试验。见附件。

Testing/签名 朱强敏

3. Mechanical strength&Durability will be influenced? / 产品可靠性、鲁棒性影响?

☐ yes/是 ☒ no/否

If Yes, How to evaluate? Plan? 如果是, 请说明影响和如何评估、计划?
(例如:仿真分析等)

NA

Validation/签名

4. Others component will be influenced? / 影响其它零部件?

☒ no/否

If yes, Which components will change in parallel? 如果影响, 哪些零部件也需要同时变更?

5. Manufactory/assembly/testing will be influenced? / 加工、装配、测试是否影响?

Feedback from manufacturing engineer? 生产制造装配是否影响, 反馈?

测试线需要更改设定参数, 1.功能测试PMB进气温度判断标准更改, 2.充氮压力设置需要更改。

Influence on project&product delivery time? 对项目&产品交货时间的影响?

无

Influence on manufacturing cost / 对生产成本的影响?

☐ Increase/增加, 金额 ☐ Decrease/降低, 金额 ☒ No change/不变

Production/签名 Li he

6. Influence on supplier part? Feedback from Supplier/Purchasing?

☒ NA(新零件放行,不涉及库存)

Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier?
变更后的零件, 生产可行性供应商是否确认? 反馈?

Influence on purchasing part delivery time? 对采购件交货时间的影响?

涉及变更的零件, 博世已出PR, 供应商还未来得及生产的订单, 是否已经通知供应商取消。

☐ no/否 ☐ yes/是

PPS for purchase parts/签名
(Below only for DCAC)
DCAC Mech. 签名
DCAC软/硬件 签名
Quality: 签名

7: Does the change influence on purchasing cost? 变更对采购成本的影响?

☐ Increase/增加, 金额 ☐ Decrease/降低, 金额 ☒ no change/不变

Quality: 签名

8. Changed parts Stock& treatment / 需要变更零件的库存统计和处理?
(只针对涉及到已经生产过的零件变更, 新放行零件和本条不相关)

☐ 涉及到已经生产过的零件变更

RBCW仓库和RBCD仓库, 是否有库存,

☐ 无库存 ☒ 有库存, 库存数量 允许混合发货

供应商处是否有库存,

☒ 无库存 ☐ 有库存, 库存数量

在途, 运输中是否有库存

☒ 无库存 ☐ 有库存, 库存数量

被影响到的实物库存处理指示:
(设计工程师组织会议讨论)

☒ 可以继续使用完 ☐ 改制后使用 ☐ 报废

☐ 改制费用 ☐ 报废成本

PPS / 签名 Meng Fanlong
PPS for purchase parts/ 签名
Quality for DCAC:/ 签名

9. DFMA & Sample production OPL?

☒ no/否 ☐ yes/是

DFMA 会议纪要, 请附件说明; 如果变更无需DFMA或前期PDECR已经做过DFMA, 请说明并列出之前的PD_ECR号 Any OPL not closed from last time sample production? 上次样件生产是否仍有OPL没有完成? 请说明或者附件说明

10. PM alignment? (Influence on project Timeline? Cost?):
是否同意对项目时间和成本的影响

需求样品订单数量

PM/签名 徐光 11.6.

11. 如果涉及以下情况需TCR3 审核确认。 不涉及则 ☒ N.A

☐ 库存报废, 金额 ☐ 采购成本增加, 金额 ☐ 生产成本增加, 金额

Step 3.2 Quality Assurance Items / 检查项目

Validations 验证	Plan finish date 计划完成日期	Resp.person 负责人	Comments 备注	
☐ Capability Studies_ CMK			NA	QMM/签名 Zou Lizhu
☐ Capability Studies_ MSA			NA	
☐ MAE release			NA	
☐ Cleaness test			NA	
☐ QZ test			NA	
☐ 200h PDL			NA	
☒ Test program or fixture etc.			功能合架进气温度范围需要更新	

Step 3.3 Affected documents Check/ 影响文档检查

1 Interface FMEA-relevant / IFMEA:	☐ no/否 ☒ yes/是	Resp.person: Fan Change	Due date: 2025.11.07
2.Product FMEA-relevant / DFMEA:	☒ no/否 ☐ yes/是	Resp.person:	Due date:
3. Special Charecteristics relevant/PSC:	☐ no/否 ☒ yes/是	Resp.person:	Due date:
4. IMDS relevant:	☒ no/否 ☐ yes/是	Resp.person:	Due date:
5 Offer drawing relevant:	☐ no/否 ☒ yes/是	Resp.person: Fan Change/Li Pingzheng/Wang Xiaolong	Due date: 2025.11.07
6. TCD relevant:	☐ no/否 ☒ yes/是	Resp.person: Qian Jiong/Sun Xudong	Due date: 2025.11.15
7. Norm, WB, HF... relevant:	☐ no/否 ☒ yes/是	Resp.person: Wu Zhaohui	Due date: 2025.11.10

Step 4 Technical feasibility & Validation plan approval / 技术可行性&验证计划批准

Design team leader	Development SM	SMP-TMS	RBCW / TCR3
Signature / 签名:	Signature / 签名:	Signature / 签名:	Signature / 签名:
康强 2025.11.06	陈树 2025.11.26		2025 谢友福

Development Changes in PD-Phase

产品开发更改

Edition 06

Doc. Approval:
RBCD/EIS
Effective date:20250610

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DC No. / 开发更改编号: PDECR_0053
Date /日期: 20251105

Step 5 Documents release / 文档发布

1. Documents release (drawing, offer drawing, Bom, Spec.,) / 文档发布 (图纸, 客户图纸, ...)

Development confirmation/ 研发确认: _____

Date /日期:

Step 6.1 Implementation check list / 导入清单

Departments	Y/N	Description	Responsible	Due date
Development	N	Change BOMs & Drawings & Documents in POE system		
	Y	Inform documents update (check work-on can met requirements)	All designe	2025.11.07
	Y	Update Offer drawing, TCD, D-FMEA	All designe	2025.11.07
	N	Norm, WB, HF...		
	N	MoC, IMDS.		
Manufacturing	N	Related (Production/Testing) equipement be ready on site		
	Y	Related (Production/Testing) program be ready	Wu Zhaohui	2025.11.10
	N	Related (Production/Testing) tooling / cutting / fixture etc. be ready		
	N	Old tooling / cutting / fixture disposal		
	N	Old materials disposal		
	N	Planner update the planning sheet		
	N	Update FMEA		
	N	Update CP/FC (Control Plan/Flow Chart)		
	Y	Update WI/PDS (Include attachments.)--》测试记录表	Wu Zhaohui	2025.11.10
	N	First batch Mark, Special Mark (Inside Package)		
	N	First batch Mark, Special Mark (Outside Package)		
COS	Y	Training	Wu Zhaohui	2025.11.10
	N	Confirm the storage of old parts and coordinate the introduction date for new part (RM)		
	N	Confirm the delivery date of old parts and first delivery of new parts (FG)		
	N	Check sample orders which affected:material order of CKD		
	N	Confirm production scheduling according to the alignment, any changes share the information to MOEx,MFE		
	N	Confirm the old stock/ do prioritize delivery and inventory handling		
PPS (PPS->PMQ/PQA)	N	Inform the first delivery to PMO		
	N	Check sample orders which affected: material order of purchasing		
	N	Inform internal related departments(COS,MFE,MOEx)with		
QMM1	N	Update incoming inspection plan		
	N	Update testing programe on testing equipment		
EIS4	N	Update inspection plan for CKD parts		
	Y	Distribute the Offer drawing, TCD to customer	Qian Jiong	2025.11.07
LOP	N	Check 10 digit material order		
PMO	N	Check sample orders which affected: Customer order		
	N	Inform Customer the first delivery information		

Step 6.2 Implementation date / 执行日期

Planned implementation date:

变更计划执行日期 2025.11.07

充氮压力试验 不同压力下保护帽弹开状态确认

重要备注: Offer drawing中要求压缩机拧松保护帽开启使用时需要能感受到内部正压力。

- **Background:** 产线和质量部反馈当前设定压力过高，在拆保护帽时，保护帽冲击力较大。
- 基于不同压力下的试验结果，决定降低充氮压力：0.15MPaA~0.25MPaA → 0.13MPaA~0.19MPaA。

0.112MPaA



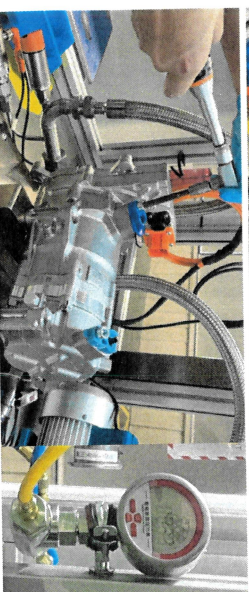
拆保护帽时无明显压力释放声，只在拔出时有“呲”声



0.13MPaA



拆保护帽时能听到压力释放声，且冲击力很小



0.15MPaA



拆保护帽时能听到压力释放声，冲击力不大



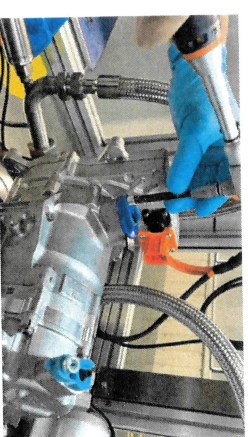
Internal C-SCI | RBC/DEIS
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0.19MPaA



拆保护帽时能听到压力释放声，冲击力不大



0.25MPaA



冲击力大，需挡板保护

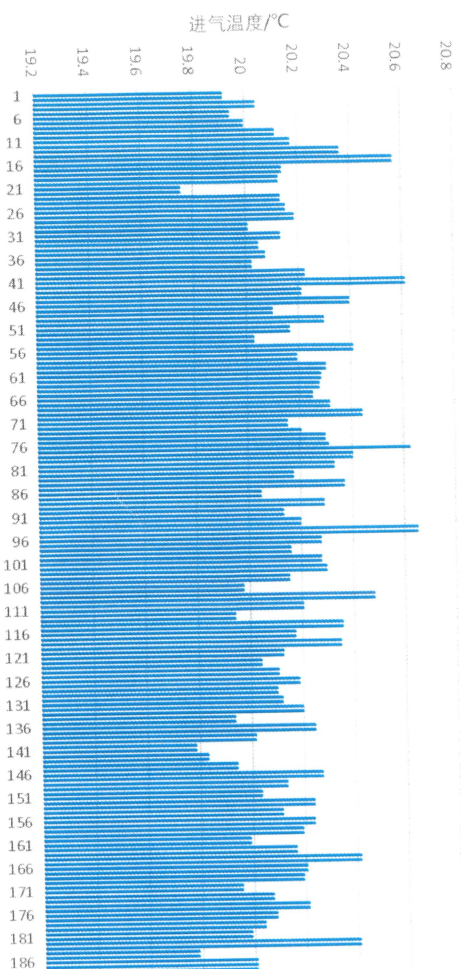


结果: 压力在0.13MPaA~0.19MPaA之间，能够在拧松保护帽过程中感受到压缩机内正压力，并且其冲击力也不大。

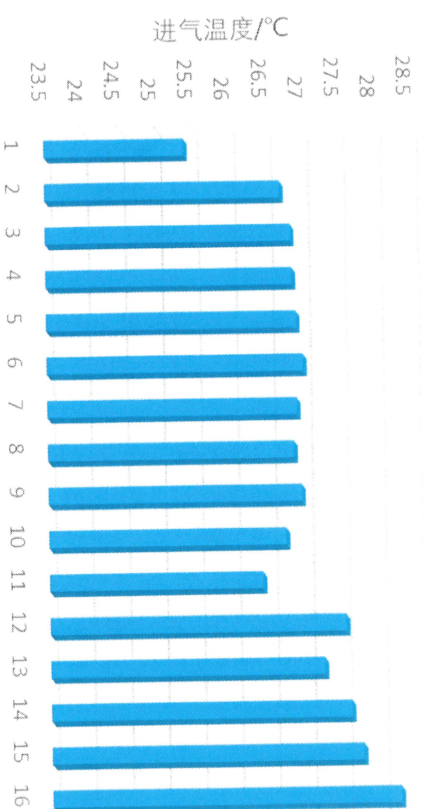
eCOMP气测台架进口温度 台架进口温度传感器记录数据

- ▶ 下线气测PMB中进气温度定义范围： $24.0\pm3^{\circ}\text{C}$ 。由规范定义初期为数不多的几台压缩机数据确定了初始值。测试压缩空气由FCM集中供气，空气温度受季节影响较大，当前气测台架只有气体干燥作用，无控温作用。基于过去将近一年的实际生产数据，调整进气温度设定至 $25.0\pm6^{\circ}\text{C}$ ，对压缩机下线功能检查无影响。

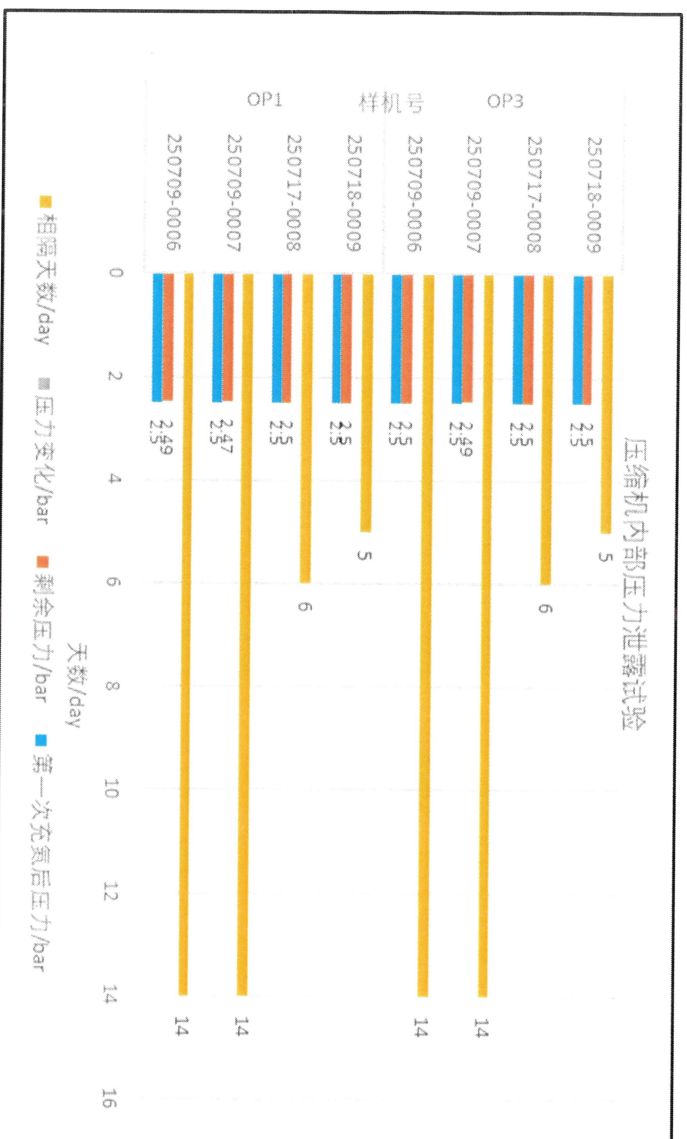
12月份进气温度监控187组测试数据@57CC



八月份进气温度监控16台数据@45CC



充氮压力试验 压缩机内部压力泄露试验



结果: 压缩机内部2.5bar压力下半个月内几乎无泄露, 说明当前保护帽装配状态下内部压力保压能力良好。

理论上若压力小于2.5bar, 泄露率会更低。